Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 11

Roll No : 496

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Progrmas based on Python Data Structure + UDF + File Handling

Practical 1: Count Lines, Words and Characters

Problem Definition:

Write a Python program to read text from a file and count the total number of lines, words, and characters. The result should be written to an output file.

In [23]:
```
fp = open("../lab 10/new2.txt" , "r")

lines = fp.readlines()
fp.close()
print("Total lines :" , len(lines))

word_count = 0
char_count = 0
for i in lines:
    words = i.split()
    word_count  += len(words)
    char_count += len(i)

print("Total words :",word_count)
print("Total characters :",char_count)

fp = open("file1.txt" , "a+")
fp.write("program 1 :\n")
fp.writelines(["lines :",str(len(lines)) ,"\nwords :", str(word_count) , "\ncharacters :",str(char_count)])
fp.close()
```

```
Total lines : 10
Total words : 166
Total characters : 1090
```

Practical 2: Search a Word in a File

Problem Definition:

Write a Python program to search a given word in a text file and count its occurrences. Store the result in an output file.

In [37]:
```
def findword(lines , word):
    for i in lines:
        # if (i.find(word) != -1):
        if (word in i):
            return True
    else:
        return False

def countitems(lines ,word):
```

```
    count = 0
    for i in lines:
        # if (i.find(word) != -1):
        if (word in i):
            count += i.count(word)
    return count

word = input("Enter word to find in file :")
fp = open("../lab 10/new2.txt" , "r")
lines = fp.readlines()
fp.close()

fp = open("file1.txt" , "a+")
fp.write("\nprogram 2 :\n")
if findword(lines , word) and countitems(lines , word) != 0:
    fp.writelines(["word :" , word , "\nexist :" ,str(True) , "\nCount :" , str(countitems(lines , word)) ])
    print("word found")
else:
    fp.writelines(["word :" , word , "\nexist :" ,str(False)])
    print("word not found")
fp.close()
```

```
word not found
```

Practical 3: Student Marks and Average

Problem Definition:

Write a Python program to read roll number, name, and marks of students from a file. Store marks in a list, calculate the average using a user-defined function, and write student details along with average marks to an output file.

In [50]:

```
def avgcal(marks):
    return sum(marks) / len(marks)

fp = open("StudentDetail.txt", "r")
lines = fp.readlines()
fp.close()

fp = open("file1.txt", "a+")
```

```
fp.write("program 3 :")

for i in lines:
    line = i.split()
    print(line)
    marks = list(map(int, line[2].split(',')))
    print(marks)
    fp.writelines([
        "\nRoll No.: ", line[0],
        "\nName: ", line[1],
        "\nAverage marks: ", str(avgcal(marks)),
        "\n"
    ])

fp.close()
```

```
['101', 'dhruv', '78,67,63,89,83']
[78, 67, 63, 89, 83]
['102', 'rajarshi', '56,68,46,67,75']
[56, 68, 46, 67, 75]
['103', 'labdhi', '45,36,38,43,42']
[45, 36, 38, 43, 42]
['104', 'archana', '65,67,87,66,89']
[65, 67, 87, 66, 89]
['105', 'rucha', '34,33,36,37,41']
[34, 33, 36, 37, 41]
['106', 'jeel', '76,79,87,56,65']
[76, 79, 87, 56, 65]
```

Practical 4: Maximum and Minimum using Tuple

Problem Definition:

Write a Python program to read numbers from a file, store them in a tuple, and find the maximum and minimum values. Write the result to an output file.

In [4]:
```
fp = open("numbers.txt","r")
nums = tuple(map(int,fp.read().split()))
fp.close()
```

```
mx = max(nums)
mn = min(nums)

print(mx , mn)

fp = open("file1.txt","a")

fp.write("\nprogram 4 :\n")
fp.write("max :" + str(mx) + "\n")
fp.write("min :" + str(mn) + "\n")

fp.close()
```

```
89 10
```

Practical 5: Remove Duplicate Words using Set

Problem Definition:

Write a Python program to read words from a file and remove duplicate words using a set. Store the unique words in an output file.

In [5]:
```
fp = open("words.txt","r")
words = fp.read().split()
fp.close()

unique = set(words)

fp = open("file1.txt","a")

fp.write("\nprogram 5 :\n")

for w in unique:
    fp.write(w + " ")

fp.write("\n")
fp.close()
```

Practical 6: Word Frequency using Dictionary

Problem Definition:

Write a Python program to count the frequency of each word in a text file using a dictionary and write the result to an output file.

In [7]:
```
fp = open("freq.txt","r")
words = fp.read().split()
fp.close()

freq = {}

for w in words:
    if w in freq:
        freq[w] += 1
    else:
        freq[w] = 1

fp = open("file1.txt","a")

fp.write("\nprogram 6 :\n")

for k,v in freq.items():
    fp.write(k + " :" + str(v) + "\n")

fp.close()
```

Practical 7: Student Result and Grade Calculation

Problem Definition:

Write a Python program to read roll number, name, and marks of students from a file. Use a user-defined function to determine grade (A/B/C/Fail) and write the student result to an output file.

Grade Criteria:

- Marks ≥ 75 : Grade A
- Marks ≥ 60 : Grade B
- Marks ≥ 40 : Grade C
- Marks < 40 : Fail

In [9]:
```
def grade(avg):
    if avg >= 80:
        return "A"
    elif avg >= 60:
        return "B"
    elif avg >= 40:
        return "C"
    else:
        return "F"

fp = open("StudentDetail.txt","r")

fp1 = open("file1.txt","a")
fp1.write("\nprogram 7 :\n")

for line in fp:
    data = line.split()

    roll = data[0]
    name = data[1]

    marks = list(map(int,data[2].split(",")))

    avg = sum(marks)/len(marks)

    g = grade(avg)

    fp1.write(roll + " " + name + " grade :" + g + "\n")

fp.close()
fp1.close()
```

In [ ]: